

The Job Sequencing and Tool Switching Problem

Consolidated Open Problems and Their Current Status

This is an index of the open problems raised across Parts I–VII, with precise statements, current status, and pointers. “Verified” means checked by the scripts in `plans-genai/_verification/` (see `VERIFIED_FACTS.md`). Conjectures are labelled as such; nothing here is claimed as proved unless it is proved in the cited Part.

Gap and Approximation Ratio (Part V)

OP 1 (Tight ratio for $b = 3$; conjecture). Is $H/Z_{\text{SSP}}^* \leq 4/3$ for every $b = 3$ instance, with equality only at the 6-job ring? *Status:* open. Strong evidence — no counterexample in 10691 exhaustively enumerated $b = 3$ edge instances, nor in the copy-paste or ring families (Part V). The earlier route via $\sum_k R_k \leq H/4$ is *false* (counterexample $\{a, b, c\}, \{a, b, d\}, \{c, d, e\}$) and has been retracted.

OP 2 (Extremal ratio for general b). What is $\sup_I H/Z_{\text{SSP}}^*$ as a function of b ? *Status:* open. Verified data: $b = 3$ attains $4/3$; the $b = 4$ two-tool family attains only 1, while a $b = 4$ three-tool sliding ring attains $6/5$. The dependence on b (and on tool-set sizes) is not characterised.

OP 3 (Structural characterisation of zero gap). Give a necessary-and-sufficient structural condition for $H = Z_{\text{SSP}}^*$. *Status:* open. Two *sufficient* conditions are proved (the heuristic meets either lower bound $K^* - 1$ or $|U| - b$; Part V); and empirically every positive-gap $b = 3$ edge instance has $K^* = 3$. A full characterisation (e.g. via boundary tool-overlap / hypergraph structure) is missing.

Grouping Selection (Parts V–VI)

OP 4 (Joint grouping–sequencing). The JGP+GSP heuristic fixes a JGP-*optimal* grouping ($K = K^*$). It is *proved* (Part V) that a sub-optimal grouping ($K > K^*$) can strictly beat it — on the 6-ring a $K = 4$ grouping attains $Z_{\text{SSP}}^* = 3 < 4 = H$. The open problem is to characterise, or efficiently optimise over, the joint choice of grouping and sequencing; the relevant objective is the joint cost, not K^* .

OP 5 (Beyond one-directional lookahead). The MWHP ordering result (Part VI) is exact only for the one-directional lookahead policy and coincides with the optimum at zero slack; with positive slack, bidirectional padding can be strictly cheaper (the sliding-window family has true cost $|U| - b = (K-1)(r-\text{ov}) - (b-r)$, below the MWHP value $(K-1)(r-\text{ov})$). A polynomial exact GSP for positive-slack groupings, or a proof of NP-hardness, is open.

Exact Formulations (Parts II, IV)

OP 6 (The exchange argument; central to Part IV). Does every uniform-SSP instance admit a *cluster-simple* optimal configuration path (each ordered cluster pair traversed at most once)? If yes, the cluster-level MTZ formulation is exact; if no, it is only a restriction (heuristic upper bound). *Status:* open. Proved for pairwise-incompatible instances and the 6-ring (Part IV).

OP 7 (LP strength and net solver benefit of MTZ). The cluster-MTZ relaxation is no tighter than SEC and is weakened further by aggregation (Part IV). Quantify the LP gap on SSP instances (with and without Desrochers–Laporte lifting), and measure whether removing row generation yields a net branch-and-price speed-up versus the SEC and SSPMF formulations on the Catanzaro Tabela1C set. *Status:* open, computational.

OP 8 (Fractional Benders cuts). The branch-and-Benders solver currently adds optimality cuts only at integer master solutions (KTNS subproblem). Deriving valid cuts at *fractional* master solutions from a relaxation of the KTNS dual — enabling user cuts that tighten the bound before branching — is open.

Cost-Function Variants (Part VII)

OP 9 (Tight collapse condition). The conditions (C1)+(C2)+(C3) are sufficient for collapse of an SSP variant to the JGP; the necessity analysis (Part VII) shows (C3) necessary and (C1),(C2) dispensable. A single necessary-and-sufficient characterisation of collapse, beyond the reduction to the threshold $\rho_c(I)$, is open.

OP 10 (Setup-time regime in practice). For the magazine/tool setup-time ratio ρ , the TD-SSP approximation ratio $R(\rho)$ decreases from $\leq 4/3$ at $\rho = 0$ to 1 as $\rho \rightarrow \infty$, with ring threshold $\rho_c = 1$ (Part VII). Whether the manufacturing regime $\rho \in [3, 60]$ (Privault–Finke) makes JGP+GSP exactly optimal on broader instance classes (not just ring-like) is open.

Cross-references: OP 1–4 (Part V, 05); OP 5 (Part VI, 06); OP 6–7 (Part IV, 04); OP 8 (BBC code); OP 9–10 (Part VII, 07).